

Finly: AI Explains the Trade. Code Controls the Money.

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Finly is a paper-trading bot that lets AI study the market without giving it control of the account. Paper trading follows real prices with virtual money. In Finly's first session on a dedicated \$100,000 Alpaca paper account, the portfolio gained \$95.32 while SPY—the fund commonly used to represent the S&P 500—lost \$57.99 from the same starting point. Finly therefore finished the session \$153.31 ahead. Alpaca recorded fifteen completed buy-or-sell fills as the portfolio was assembled, with no deposits, withdrawals, or options held at that close [1]. One session cannot establish future performance, but it does show that Finly moved from an idea to a running system with a result that judges can inspect.

Finly is one product built around a clear division of responsibility. Its rules-based portfolio keeps 48.5 percent in QQQ, which tracks the Nasdaq-100; divides another 48.5 percent among three sector funds with stronger longer-term price trends; and holds 3 percent in cash. When the evidence supports it, Finly can also consider one small trade tied to SPY using options—time-limited contracts whose value depends on SPY's price. A hosted AI model (Qwen3-32B) reads public Alpaca news and explains what supports the trade, what argues against it, and what remains uncertain. The model does not choose how much money to risk or write the broker order. Fixed, tested code chooses the direction, contracts, quantity, maximum loss, and every field sent to Alpaca. The competition limit is \$500 of maximum possible loss per options trade. AI can make Finly more cautious or stop a trade; it cannot raise that limit. We fixed the portfolio and competition rules before the paper test began, so they could not be revised after seeing the result [2].

Every options decision follows the same visible path. Finly records current fund prices, current prices for the options, trading activity, and public Alpaca news. The model explains the evidence; code builds an exact pair of options and checks that the data are current, the contracts are actively traded, the possible gain and loss are calculated correctly, and the virtual account has enough capacity. Finly then repeats the decision after removing information sources and changing important inputs. A failed check means no trade. A successful approval is tied to one exact order, sent through Alpaca's official connection, and checked against the paper account afterward. The cloud runner saves its progress before each action, so a restart cannot silently create a second order. The public dashboard reports a safe account summary and the saved reason for trading—or staying out [1,3].

The historical evidence provides a longer test of the portfolio rule. In a January 2013–August 2026 simulation after modeled trading costs, \$10,000 became \$106,711 with Finly and \$68,082 with SPY, a \$38,629 difference in ending wealth. A simpler version built from long-running industry data was tested on 21,218 market days from 1927 through 2007. It averaged 13.37 percent growth per year versus 9.48 percent for the market, remained ahead in all twenty-one tests that changed the monthly update day, and stayed ahead when the assumed trading cost increased fivefold. The interactive options demonstration made the risk controls equally concrete: one illustrative SPY plan had a \$366 maximum loss and \$634 maximum gain. Finly kept the same decision after removing each of four sources in turn and after thirty-two small input changes; when the evidence conflicted, its checks did not authorize a trade. No broker order or fill occurred in the demonstration [2]. The public verification run found 809 automated tests: 807 passed, none failed, and two were skipped [4].

Finly does not try to win by asking an AI to sound more certain. It makes every investment idea pass through software that can explain the evidence, limit the loss, and leave the account untouched. That distinction is useful to an ordinary visitor, who can see what Finly bought and whether it made money, and to a technical judge, who can trace the same decision through the model output, risk calculation, broker order, and account read-back. The project satisfies the hackathon's options requirement while using Alpaca's official infrastructure, a dedicated competition account, meaningful AI analysis, explicit risk checks, and visible paper-account results. Judges can open the working application, replay both an approval and a safety stop, inspect the public repository, reproduce the historical tests, and follow the mathematical and engineering arguments in the linked technical documents [5].

References

[1] Finly, [first-close measurement](#), [competition deployment record](#), and [latest options decision receipt](#). [2] Finly, [frozen competition protocol](#), [quantitative evidence](#), and [earlier-era replay](#). [3] Alpaca, [official MCP server](#); Finly, [cloud runner design](#). [4] Finly, [public automated-test run](#). [5] Finly, [live application](#), [public repository](#), [mathematical technical note](#), [engineering appendix](#), [demo video](#), and [pitch deck](#).